Representative Structural Calculation Package

Regular office frame

Item	Value
Project	Regular office frame
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	01-regular-office-frame

Nodes	Members	Loads	Combos	Max disp.	Max util.
45	**84**	**240**	**28**	**5.546 mm**	**0.381**

Model And Design Basis

Item	Value
Building type	office
Description	Four-story regular moment frame used as a baseline office building.
Occupancy	Office
Bounds X/Y/Z	12.00 / 10.00 / 14.40
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	48	0.000 kN	0.000 kN	-2016 kN
L	live	48	0.000 kN	0.000 kN	-840.00 kN
WX	wind	36	64.80 kN	0.000 kN	0.000 kN
WY	wind	36	0.000 kN	77.76 kN	0.000 kN
EX	seismic	36	175.85 kN	0.000 kN	0.000 kN
EY	seismic	36	0.000 kN	175.85 kN	0.000 kN

Load Derivation

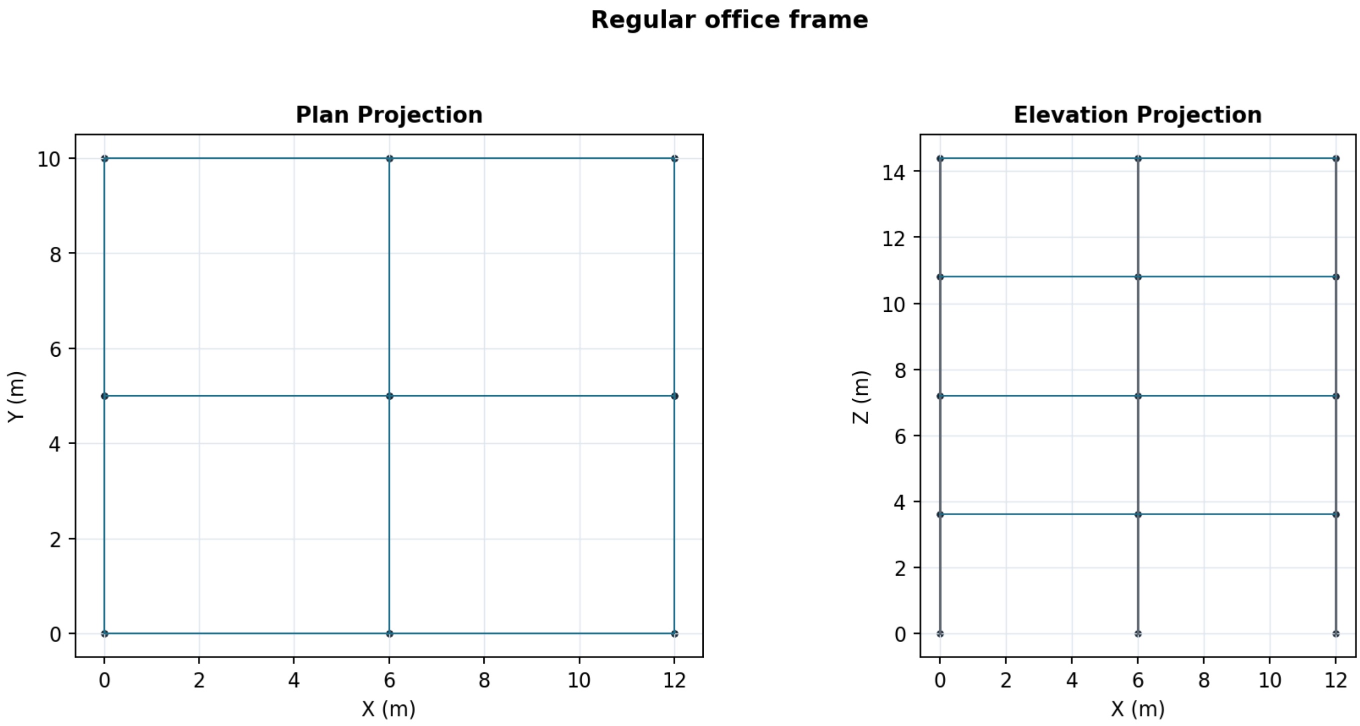
Story	Area	D total	L total	Wind X	Wind Y
1	120.00	504.00 kN	240.00 kN	16.20 kN	19.44 kN
2	120.00	504.00 kN	240.00 kN	16.20 kN	19.44 kN
3	120.00	504.00 kN	240.00 kN	16.20 kN	19.44 kN
4	120.00	504.00 kN	120.00 kN	16.20 kN	19.44 kN

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required

Preset	Status	Missing	Missing any	Generated
KDS-ST-03	blocked	-	Lr, S, R	0

Geometry



Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	1.135 mm	0.124	0.000
KDS-ST-02	OK	1.399 mm	0.165	0.000
KDS-ST-04-WX-P	OK	2.049 mm	0.182	0.000
KDS-ST-04-WX-N	OK	2.049 mm	0.182	0.000
KDS-ST-04-WY-P	OK	2.169 mm	0.188	0.000
KDS-ST-04-WY-N	OK	2.169 mm	0.188	0.000
KDS-ST-05-EX-P	OK	5.546 mm	0.252	0.000
KDS-ST-05-EX-N	OK	5.546 mm	0.252	0.000
KDS-ST-05-EY-P	OK	5.071 mm	0.249	0.000
KDS-ST-05-EY-N	OK	5.071 mm	0.249	0.000
KDS-ST-06-WX-P	OK	1.786 mm	0.119	0.000
KDS-ST-06-WX-N	OK	1.786 mm	0.119	0.000
KDS-ST-06-WY-P	OK	1.924 mm	0.125	0.000
KDS-ST-06-WY-N	OK	1.924 mm	0.125	0.000
KDS-ST-07-EX-P	OK	5.453 mm	0.189	0.000
KDS-ST-07-EX-N	OK	5.453 mm	0.189	0.000

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M32	OK	0.381	steel-deflection	KDS-ST-05-EX-P
2	M5	OK	0.380	steel-interaction	KDS-ST-05-EX-P
3	M33	OK	0.378	steel-deflection	KDS-ST-05-EY-N
4	M31	OK	0.378	steel-deflection	KDS-ST-05-EY-P
5	M35	OK	0.378	steel-deflection	KDS-ST-05-EX-P
6	M29	OK	0.378	steel-deflection	KDS-ST-05-EX-N
7	M34	OK	0.377	steel-deflection	KDS-ST-05-EX-N
8	M36	OK	0.377	steel-deflection	KDS-ST-05-EX-P
9	M28	OK	0.377	steel-deflection	KDS-ST-05-EX-N
10	M30	OK	0.377	steel-deflection	KDS-ST-05-EX-P
11	M2	OK	0.338	steel-interaction	KDS-ST-05-EX-P
12	M8	OK	0.338	steel-interaction	KDS-ST-05-EX-P
13	M4	OK	0.333	steel-interaction	KDS-ST-05-EY-N
14	M6	OK	0.333	steel-interaction	KDS-ST-05-EY-P
15	M23	OK	0.318	steel-deflection	KDS-ST-05-EX-N
16	M24	OK	0.316	steel-deflection	KDS-ST-05-EY-P
17	M22	OK	0.316	steel-deflection	KDS-ST-05-EY-N
18	M20	OK	0.316	steel-deflection	KDS-ST-05-EX-P
19	M26	OK	0.316	steel-deflection	KDS-ST-05-EX-N
20	M19	OK	0.315	steel-deflection	KDS-ST-05-EY-N

Member Design Trace Matrix

Item	Value
Checked members	84
Unimplemented members	0
NG / WARN	0/0
Max utilization	0.381

Member	Type	Status	Util.	Formula	Action
M32	steel	OK	0.381	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M5	steel	OK	0.380	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M33	steel	OK	0.378	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M31	steel	OK	0.378	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M35	steel	OK	0.378	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

M29	steel	OK	0.378	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M34	steel	OK	0.377	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M36	steel	OK	0.377	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M28	steel	OK	0.377	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M30	steel	OK	0.377	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M2	steel	OK	0.338	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M8	steel	OK	0.338	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M4	steel	OK	0.333	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M6	steel	OK	0.333	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M23	steel	OK	0.318	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M24	steel	OK	0.316	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M22	steel	OK	0.316	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M20	steel	OK	0.316	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M26	steel	OK	0.316	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M19	steel	OK	0.315	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	84	0	0.381
Connections	84	23	2.561
Foundations	9	23	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.
- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.